

Ganesh Ravichandran

US Citizen, authorized to work in the US for any employer

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SUMMARY

Infrastructure engineer who thrives at the intersection of rapid experimentation, performance optimization, and production-scale ML systems.

Notable accomplishments from six years at early-stage startups:

- Delivered a high-throughput ML pipeline leveraging embedding generation, dimensionality reduction, and vector search processing for 50,000+ files per week, with incremental learning and intelligent checkpointing.
- Built a static file serving pipeline that achieved sub-100ms latency serving 300,000+ files per week, with 4-5x bandwidth reduction through compression techniques.
- Led a cross-functional team of five engineers through a rapid product pivot under tight time and budget constraints, maintaining an aggressive experimentation pace that cut the expected timeline in half and delivered a beta launch within five weeks.

I approach complex systems by understanding their fundamental principles with deep curiosity, which has developed my ability to rapidly master new domains, think creatively about technical challenges, and deliver outsized engineering impact. Following my last role, I have been pursuing an intensive technical deep-dive into ML systems engineering, GPU architecture, and LLM training/inference optimization.

EDUCATION AND SKILLS

Columbia University, B.A. in Computer Science

Sep. 2014 – Feb. 2018

Proficient: Python, NumPy, TypeScript/JavaScript, Kubernetes, Docker, AWS

Familiar: PyTorch, JAX, CUDA, C++

EXPERIENCE

Independent ML Performance and Systems Research

Oct. 2024 – Present

Conducted self-directed research into GPU architectures, Transformer optimization, and distributed training methodologies to advance understanding of ML performance bottlenecks and scalability challenges. Published in-depth technical articles on optimization techniques at thebatmanofbutler.substack.com.

Key projects include:

- Designing and assembling a **custom GPU cluster** with four NVIDIA GTX 1070 Ti GPUs with full $\times 16$ PCIe bandwidth and AMD Threadripper 2920X, optimizing for parallel training workloads.
- Implementing and **training GPT-2 locally**, leveraging PyTorch Memory Profiler for **performance analysis** and implementing activation recomputation to **reduce memory overhead during training**.
- Studying current literature on computational efficiency in deep learning, including data parallelism, ZeRO memory optimization, quantization techniques, and speculative decoding, to build **foundational knowledge in ML performance optimization**.

Senior Software Engineer at SOOT (AI copilot for visual media)

Feb. 2021 – Oct. 2024

First full-time engineering hire. Architected and maintained critical infrastructure for ML-driven applications. Built scalable systems handling hundreds of thousands of files per month and optimized distributed computing resources. Mentored junior engineers and drove technical architecture decisions.

Engineered high-performance ML pipeline architecture for production workloads:

- Built an ML-powered file ingestion pipeline leveraging CLIP embeddings, UMAP dimensionality reduction, FAISS vector search, and Stable Diffusion image generation. **Processed over 50,000 image uploads per week with 99.9% reliability.**

- Maintained and optimized a distributed architecture of Kubernetes worker pods and message queues to **promote throughput, minimize request latency, and handle expensive, concurrent ML workloads**. Implemented intelligent UMAP model updates with incremental learning capabilities to **maintain model quality at scale**.

Optimized distributed computing infrastructure for scale and performance:

- Designed and developed a highly-scalable static file serving pipeline using CloudFront, Lambda@Edge, and S3 to handle distributed data access at scale. Integrated multiple parts of our microservices architecture, including CloudFront, Lambda@Edge, S3, and custom RBAC logic. **Scaled system to handle over 300,000 file requests per week with sub-100ms latency**.
- Implemented horizontal autoscaling for Kubernetes clusters based on queue depth metrics from SQS, optimizing resource utilization. **Achieved 2x increase in processing capacity while reducing infrastructure costs**.
- Engineered multi-resolution image serving pipeline by leveraging texture atlas compression techniques to **reduce bandwidth usage 4-5x compared to JPEG**. Supported both low-latency previews and lazy-loaded high-quality originals to **serve 10,000+ images per client per session**.
- Developed a local filesystem synchronization engine to automatically update uploaded user data based on local file changes, with the ability to **continuously watch and update over 500,000 files per user device**. This quality-of-life feature **increased average uploaded user data by 40%**.

Architected fault-tolerant systems with comprehensive observability monitoring:

- Promoted fault-tolerance through granular retry mechanisms and partial failure handling. Implemented checkpointing to preserve successful operations during batch processing, **reducing data reprocessing by 85% during failure scenarios**.
- Architected a comprehensive observability stack using OpenTelemetry and Zipkin for distributed tracing across microservices. **Improved system observability coverage by 80% and reduced mean time to detection by 25%**.
- Securely configured an Electron app update server and CI/CD pipeline with automated code signing to continuously deliver new features to ensure reliable, continuous delivery of new features to users.

Designed high-throughput API endpoints and authentication infrastructure:

- Extended GraphQL API functionality to support multiple highly scalable requests, including the implementation of a streaming architecture to **enable users to bulk download all of their account data, typically over 100,000 images**.
- Implemented authentication and authorization infrastructure integrating multiple identity providers, creating reusable libraries for service-to-service communication. Substantially reduced dev time dedicated to new auth provider integration.
- Collaborated with the design team to develop Vue components that enhanced user communication and transparency during system failures, resulting in a **90% increase in error messaging coverage**.

Software Engineer and Technical Lead at LiveStories (SaaS platform for local governments)

Sep. 2019 – Feb. 2021

Full-stack software engineer. Led development on the MVP during mid-pandemic product pivot.

Engineered core features and accelerated product performance:

- **Developed 4/7 of the MVP's critical features**, including authentication, location-based search, and chart editing.
- Rearchitected the search function for the data library and **reduced load time by 40%**.
- Shipped an email campaign launch platform for government clients to reach businesses based on their eligibility for a given aid program, used by more than half of our clients.
- Built tabular React UIs for government clients to analyze aid eligibility for over 10,000 businesses.
- Developed and maintained UIs to empower struggling businesses to apply to more than 130 municipal and federal aid programs and gain access to **over \$600,000 in available pandemic relief**.

Orchestrated MVP Launch after strategic product redirection during COVID:

- **Managed three engineers across three continents** to develop and launch an MVP to replace the legacy app.
- Advised CEO on design/development strategy and cut sprint planning meeting lengths in half.
- Facilitated transition for existing users by integrating new MVP into the existing product UX.